Learning to Generate Text-guided 3D Canine-Object Interactions via Video Diffusion Priors (Supplementary Material)

SIYU CHEN, Beijing Institute of Technology, China
ZAN WANG, Beijing Institute of Technology, China
HAO XU, Beijing Institute of Technology, China
MINGHE ZHANG, Beijing Institute of Technology, China
MINGTAO PEI, Beijing Institute of Technology, China
WEI LIANG✉, Beijing Institute of Technology, China

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A DogML3D Dataset

The raw data are sourced from DigiDogs [Shooter et al. 2024], AnimalML3D [Yang et al. 2024] and AniMo4D [Wang et al. 2025c]. An overview of the datasets is presented in Table A1, where Num, Breed, Frame, and Text denote the number of data entries, the number of breeds, the total number of frames, and the total number of text descriptions, respectively.

Table A1. **Dataset statistics.**

Dataset	Num	Breed	Frame	Text
DigiDogs	254	8	51720	492
AnimalML3D	366	6	16432	2202
AniMo4D	6104	25	524576	21202

Correspondences between the original skeletons and target skeletons are established via interpolation, which unifies the raw joint motion data into motion data of N joints, denoted as $J_{\text{tgt}} \in \mathbb{R}^{3N}$. For each breed, one sample is first selected as the template, and the parameters β, s, t, r, θ are obtained through optimization. The shape parameters β and global scaling factor s of this template are then adopted for all dogs of the same breed, while only the parameters t, r, and θ need to be optimized. The optimization loss is defined as

$$\mathcal{L}_{\text{opt}} = \lambda_{\text{J}}\mathcal{L}_{\text{J}} + \lambda_{\text{shape}}\mathcal{L}_{\text{shape}} + \lambda_{\beta}\mathcal{L}_{\beta} + \lambda_{\theta}\mathcal{L}_{\theta} + \lambda_{\text{T}}\mathcal{L}_{\text{T}} \tag{A1}$$

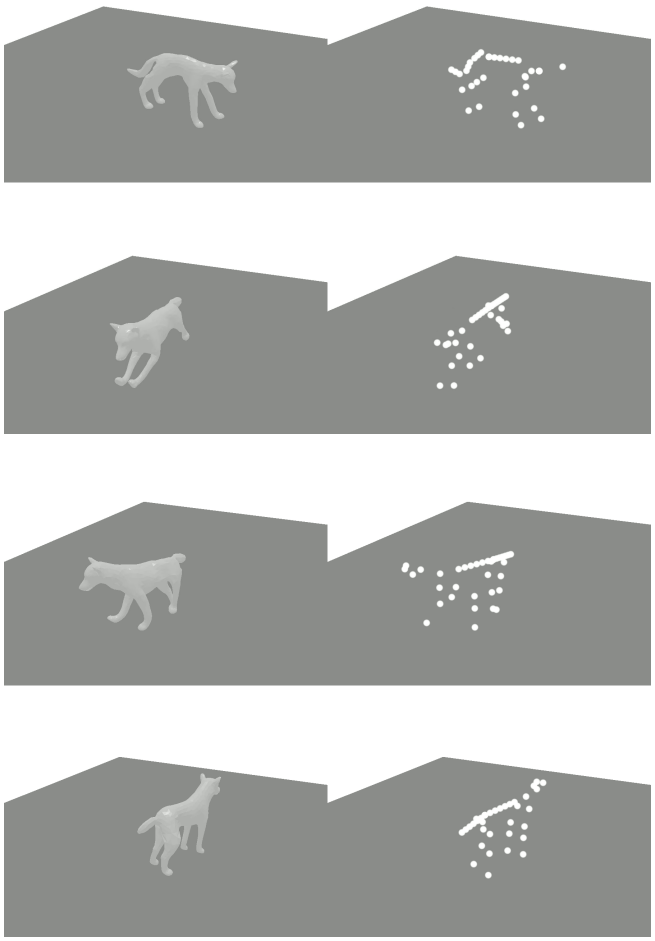


Fig. A1. **Optimization results.** The first column displays the optimized meshes and the second column shows the target joints.

Authors' Contact Information: Siyu Chen, 3120240925@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Zan Wang, wangzan@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Hao Xu, 3220251508@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Minghe Zhang, 3220251379@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Mingtao Pei, peimt@bit.edu.cn, Beijing Institute of Technology, Beijing, China; Wei Liang, liangwei@bit.edu.cn, Beijing Institute of Technology, Beijing, China.

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Fig. A2. **Overview of object sets.** Objects on the blue floor correspond to the training set, while those on the red floor belong to the held-out set.

Table A2. **Object statistics for training and held-out sets.**

Training		Held-out	
Category	Num	Category	Num
Sofa	5	Sofa (new geometry)	3
Bed	5	Bed (new geometry)	2
Chair	5	Bookshelf	2
Table	5	Dog bed	1
Bowl	3	Wooden fence	1
Stool	3	Tennis ball	1
Swimming pool	3	Water dispenser	1

where $\mathcal{L}_{\text{J}}$ denotes the L_2 loss on 3D joint locations, and $\mathcal{L}_{\text{shape}}$ denotes the shape prior loss derived from WLDO [Biggs et al. 2020] and SMALify [Biggs et al. 2018]. $\mathcal{L}_\beta$ and $\mathcal{L}_\theta$ denote the regularization losses for β and θ, respectively. $\mathcal{L}_{\text{T}}$ is the temporal smoothness loss, which is used to constrain the differences in joints and poses between adjacent frames. In implementation, the optimization process is divided into two stages, where the temporal smoothness loss is not applied in the first stage. The L-BFGS optimizer is adopted, with 800 epochs of training for the first stage and 300 epochs for the second stage. The learning rate is set to 0.01 for both stages. The hyperparameters λ_{J}, λ_{shape}, λ_β, λ_θ and λ_{T} are set to 1.0, 1.0, 1.0, 1.0 and 10.0, respectively.

After optimization, we normalize the dataset, akin to HumanML3D [Guo et al. 2022], which involves three sequential steps: (1) applying horizontal flipping to the motion sequences for increased data diversity; (2) placing the root joint of the first frame at the origin of the xz-plane and translating the motion to the ground plane; and (3) finally aligning the first frame to face the positive z-axis. We trim the number of frames to ensure that the motion sequences range from 20 to 300 frames. Partial optimization results are shown in Fig. A1, where the first column presents the optimized meshes, while the second column illustrates the target skeletons; the gray regions denote the floor.

B Object-aware Text Augmentation

Object Set. We construct an object set to support object-aware interaction modeling. The training set contains common interaction objects, while a held-out set is used to assess generalization to unseen geometries and novel categories. Tab. A2 summarizes the objects in the training and held-out sets, and Fig. A2 provides a visual overview.

Text Augmentation Prompt for LLM

You are an expert at rewriting animal motion descriptions.

Your task is: Given an original motion description and a specified object, rewrite the sentence so that the animal naturally interacts with the object.

Input
- original_text: the original motion description
- object: the given object

Output
- spatial_relation: one of ['on', 'under', 'near', 'around', 'in front of', 'behind']
- augmented_text: the rewritten sentence

Core constraints
- Preserve the original actions and their order. Do not add, remove, or reorder actions. Motion should remain temporally continuous.
- The interaction must involve only the animal and given object. Do not introduce any additional entities or environmental elements (e.g., water, snow, ground textures).
- The rewritten sentence must explicitly describe the interaction using exactly one spatial relation from the predefined set. Choose the most reasonable relation based on the motion and object.
- The description should remain concise and natural. Avoid unnecessary details, redundant phrases, or overly long sentences.

Fig. A3. **Prompt template used for LLM-based text augmentation.**

LLM-based Text Augmentation. We employ ChatGPT-4o [Achiam et al. 2023] to perform object-aware text augmentation. Each motion instance is associated with 1–3 original descriptions, and for each description, 1–3 objects are randomly sampled for augmentation.

The model is prompted to generate interaction-aware descriptions conditioned on the original text and the sampled object. The generated text is constrained to (1) preserve the original action sequence without introducing new behaviors, (2) restrict the interaction to the canine and the specified object, (3) maintain concise and temporally coherent motion descriptions, and (4) use exactly one spatial relation from a predefined set. The detailed prompt template is shown in Fig. A3.

C Implementation Details

Training Setup. In our implementation, we adopt a frozen *CLIP-ViT-B/32* [Radford et al. 2021] encoder to extract text embeddings. The Text-to-Motion Pretraining and the Object-aware Finetuning stages are trained independently to convergence using the AdamW

Method	R-Precision ↑			FID ↓	Multimodal Dist. ↓	Diversity →	MultiModality ↑
	Top 1	Top 2	Top 3				
Real	$0.2917^{\pm.008}$	$0.4618^{\pm.006}$	$0.5567^{\pm.023}$	$1.3708^{\pm.468}$	$14.6443^{\pm.129}$	$36.2613^{\pm.385}$	–
w/o video-domain SDS	$0.2458^{\pm.017}$	$\mathbf{0.3875^{\pm.017}}$	$0.5062^{\pm.034}$	$4.8599^{\pm.467}$	$\mathbf{15.2280^{\pm.263}}$	$33.0463^{\pm.581}$	$0.8281^{\pm.324}$
w/o decoupling	$0.2396^{\pm.014}$	$0.3833^{\pm.020}$	$0.4854^{\pm.035}$	$16.7865^{\pm.132}$	$15.7143^{\pm.258}$	$28.6512^{\pm.707}$	$\mathbf{1.9999^{\pm.141}}$
w/o object-aware control branch	$0.2375^{\pm.008}$	$0.3812^{\pm.006}$	$0.5012^{\pm.023}$	$7.6372^{\pm.254}$	$15.5478^{\pm.386}$	$34.9046^{\pm.291}$	$0.5947^{\pm.201}$
w/o approach	$0.2505^{\pm.015}$	$0.3872^{\pm.017}$	$\mathbf{0.5127^{\pm.020}}$	$3.8365^{\pm.300}$	$15.3020^{\pm.276}$	$32.5018^{\pm.570}$	$1.0034^{\pm.225}$
w/o physical	$0.2490^{\pm.014}$	$0.3869^{\pm.016}$	$0.5118^{\pm.022}$	$3.7100^{\pm.240}$	$15.3328^{\pm.335}$	$32.2074^{\pm.275}$	$1.0189^{\pm.226}$
Ours	$\mathbf{0.2508^{\pm.003}}$	$\mathbf{0.3875^{\pm.015}}$	$0.5125^{\pm.013}$	$\mathbf{3.6620^{\pm.286}}$	$15.2757^{\pm.481}$	$32.7650^{\pm.687}$	$1.0140^{\pm.268}$

Table A3. **Motion prior ablation.** We analyze the effect of removing video-domain SDS, world-motion decoupling, object-aware control branch design, approach loss, and all physical losses.

optimizer with fixed learning rates. Specifically, the Text-to-Motion Diffusion Model is trained with a learning rate of 1×10^{-4} for 800K steps on a single RTX 4090 GPU with a batch size of 64, while the Object-aware Finetuning stage is trained with a learning rate of 5×10^{-5} for 50K steps on 4 RTX 4090 GPUs with a batch size of 16 per GPU. During Object-aware Finetuning, the loss weighting coefficients are set to $\lambda_{\text{MSE}} = 0.1$, $\lambda_{\text{app}} = 0.03$, $\lambda_{\text{pen}} = 0.7$, $\lambda_{\text{contact}} = 0.2$, and $\lambda_{\text{SDS}} = 0.05$, balancing the magnitudes of different loss components. The video-domain SDS gradients are computed by rendering 16 uniformly sampled video frames at a resolution of 160×256, which are then upsampled to the native 320×512 resolution required by the frozen ZeroScope model. Table A4 summarizes the peak GPU memory usage and runtime for the two training stages and their corresponding inference settings.

Table A4. **Training and inference computational cost of ZeroCOI.** Inference is measured with a batch size of 1. For multi-GPU training, peak GPU memory is aggregated across all GPUs.

Mode	Peak GPU Mem.	Runtime
Training: Text-to-Motion Pretraining	8.25 GiB	1.7 days
Training: Object-aware Finetuning	54.06 GiB	7.5 days
Inference: Text-to-Motion Prior	1.23 GiB	21.74 s
Inference: Full ZeroCOI	1.98 GiB	93.56 s

Rendering Setup. In the rendering stage, we let the virtual camera smoothly follow the canine root joint over time, keeping the subject centered in the view, which is consistent with the object-centric bias of the text-to-video diffusion model used for distillation. To mitigate single-view ambiguity and encourage viewpoint-robust supervision, the camera azimuth, elevation, and distance are randomly sampled within predefined ranges, while always maintaining a look-at constraint toward the canine root.

In addition, we incorporate a ground plane into the rendered scene, providing the video diffusion model with explicit visual context of canine–ground contact. This enables the distilled semantic gradients to encode not only object interaction cues but also physically plausible motion patterns relative to the ground surface.

D Ablation Study

To analyze how different components affect the underlying motion prior, we additionally report standard text-to-motion evaluation metrics following [Guo et al. 2022], including *R-Precision* and *Multimodal Distance*, assessing the relevance between generated motions and language prompts; *FID*, evaluating the discrepancy between the distributions of generated results and the original dataset; *MultiModality*, quantifying the average variation conditioned on text descriptions; and *Diversity*, measuring the variation within generated motions. These are reported exclusively in the ablation study, since baseline methods are based on instance-level optimization or reconstruction and do not involve text–motion training, making distribution-level metrics inapplicable and not directly comparable.

The results are reported in Tab. A3. Removing video-domain SDS results in motions that remain close to the canonical distribution but exhibit mislocalized and semantically inconsistent interactions, revealing the role of video-domain SDS in grounding interaction semantics. Disabling the world & motion space decouple strategy leads to a substantial degradation in motion quality, as evidenced by a markedly increased *FID* and *Multimodal Distance*, suggesting that directly imposing interaction constraints in the motion space introduces conflicts with the canonical motion prior, resulting in unstable motions with frequent penetrations and foot sliding. Removing object-aware control branch disrupts the learned motion manifold, showing degraded text-motion alignment and increased distributional drift, which qualitatively manifests as distorted motions or excessive in-place sliding. Finally, removing the approach loss and all physical losses results in only marginal changes in these metrics, since these losses primarily affect interaction quality (e.g., spatial variability, penetration artifacts, and contact consistency) rather than the learned motion prior. The qualitative comparisons are shown in Fig. A4 and Fig. A5.

E Additional Analysis

E.1 Analysis of Video Diffusion Priors

To further clarify the role of the video diffusion prior, Fig. A6 presents representative cases illustrating how its supervision affects the generated interactions. At the semantic level, the prior helps correct object-relative spatial relationships (*e.g.*, "near" versus "on" an object) and adjust interaction-specific motion patterns (eg,

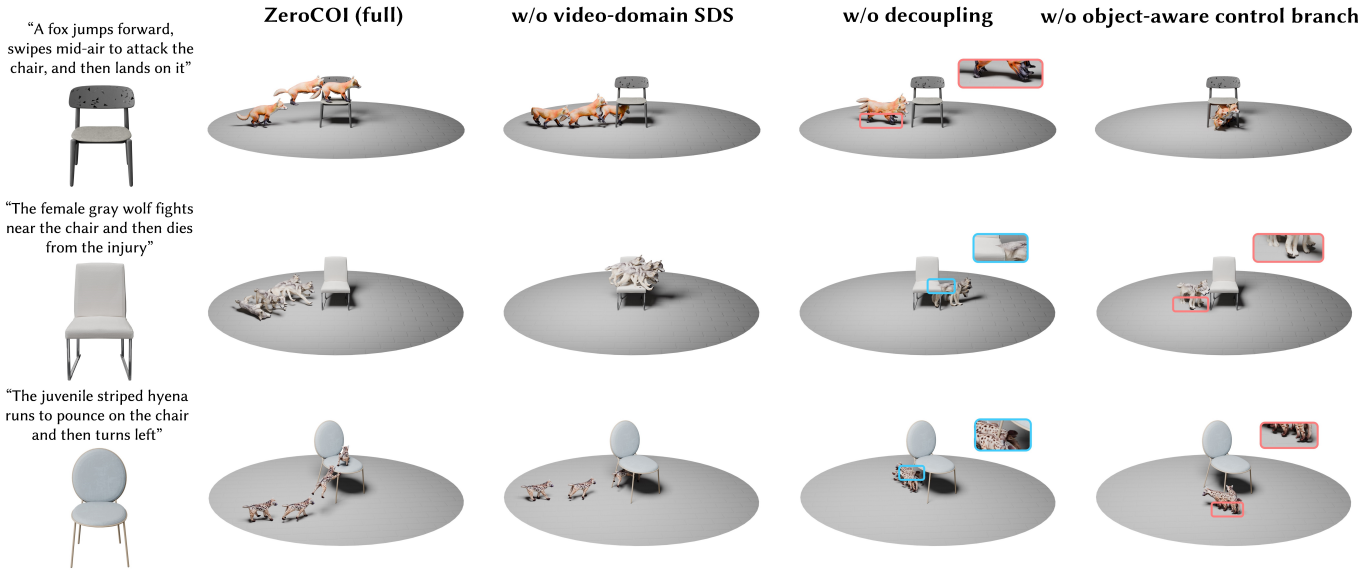


Fig. A4. **Qualitative ablation comparisons.** Ablating video-domain SDS, decoupling, or the object-aware control branch results in interactions that are semantically misaligned and physically implausible. Red indicates foot sliding, and blue denotes object penetration.

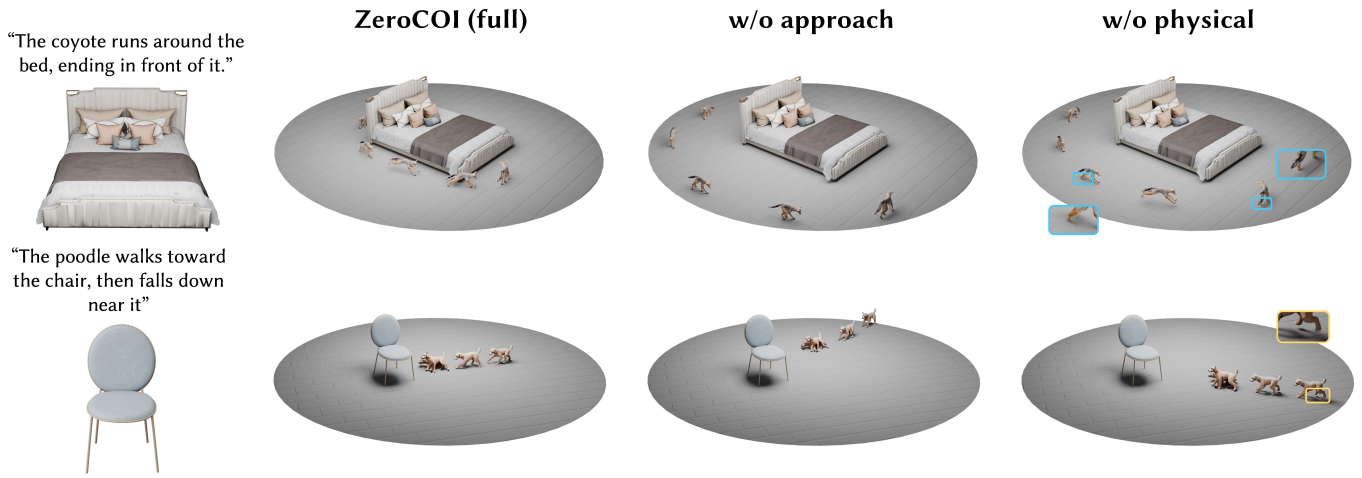


Fig. A5. **Qualitative ablation comparisons.** Removing the approach loss leads to increased spatial variability, while removing physical losses results in more minor artifacts such as floating and penetration. Yellow indicates floating, and blue denotes object penetration.

"jump onto the table" versus "jump on the ground") according to the text condition. At the physical level, it helps suppress visually salient violations (*e.g.*, large-area body-object interpenetration or obvious floating above a support surface). However, such visual-level supervision is less sensitive to subtle geometric errors (*e.g.*, small paw-surface contact gaps or minor local penetrations), which motivates the complementary use of explicit 3D physical constraints introduced in Sec. 3.5.

E.2 Analysis of Physical Fidelity

To assess how the construction of DogML3D and subsequent learning stages affect physical fidelity, Fig. A7 compares representative cases from the source skeleton motions, the retargeted DogML3D motions, the motion-prior-generated motions, and the full **ZeroCOI**-generated motions. Tab. A5 further reports the Foot Sliding (FS) and Contact metrics for these four settings, where Contact denotes the frame-averaged minimum foot-to-support-surface distance over non-jumping sequences.

Across both qualitative and quantitative comparisons, DogML3D and the pretrained motion prior exhibit comparable physical fidelity

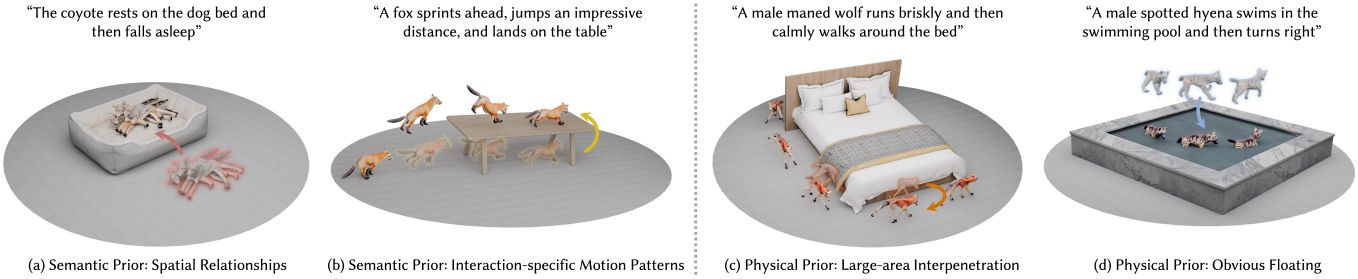


Fig. A6. **Qualitative analysis of video diffusion priors.** The semantic priors capture both spatial relationships and interaction-specific motion patterns, while the physics priors penalize visually salient physical violations such as large-area interpenetration or obvious floating.

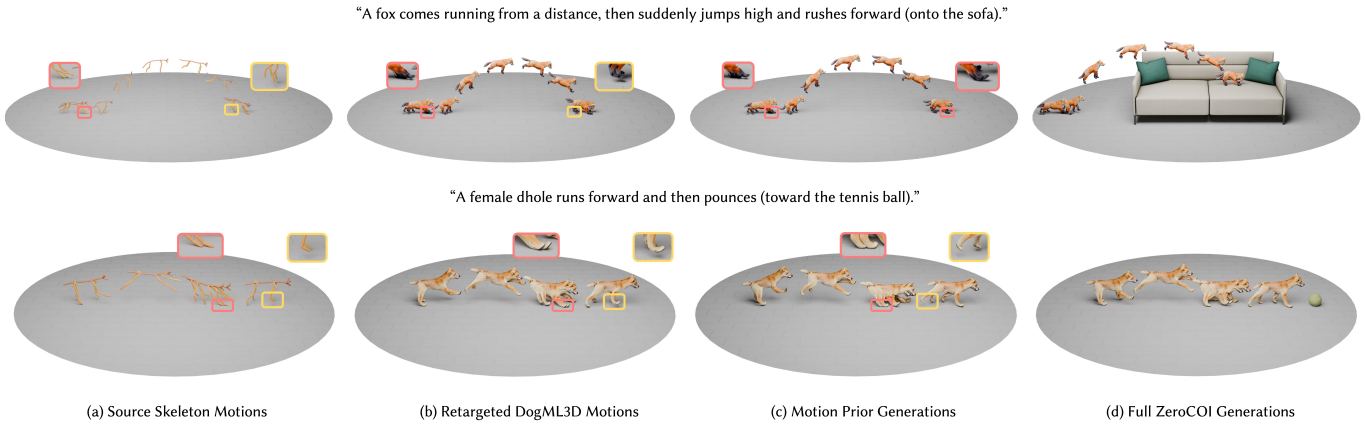


Fig. A7. **Qualitative comparison of physical fidelity.** The full **ZeroCOI** shows fewer foot-sliding and imperfect-contact artifacts while preserving natural motion. Red indicates foot sliding, and yellow indicates floating.



Fig. A8. **Compositional generalization and spatial adaptation.** (a) **ZeroCOI** composes atomic actions observed separately in the training data into coherent multi-stage interactions under text prompts. (b) **ZeroCOI** exhibits spatial adaptation, maintaining consistency between the jointly generated canonical motion and global placement under different object-relative initial configurations.

to the source motions, whereas the full **ZeroCOI** shows fewer foot-sliding and imperfect-contact artifacts and reduces both metrics. These results demonstrate that our method further improves physical plausibility beyond the underlying motion prior.

E.3 Compositional Generalization and Spatial Adaptation

Compositional generalization. As shown in Fig. A8(a), our model can compose atomic actions observed separately in the training

Table A5. **Quantitative comparison of physical fidelity.**

Setting	FS↓	Contact↓
Source Skeleton Motions	0.0964	0.0523
DogML3D	0.1035	0.0515
Pretrained Motion Prior	0.1174	0.0568
Full **ZeroCOI**	**0.0440**	**0.0291**

data into coherent multi-stage motions in response to text prompts, despite such compositions being unseen during training.

Spatial adaptation. Fig. A8(b) illustrates the consistency between the jointly generated canonical motion and global placement. When the predicted placement positions the initial canine state farther from the object, the corresponding motion includes an approach phase before the intended interaction, resulting in a longer sequence. In contrast, a closer initial state leads to a more direct interaction.

E.4 Temporal Dynamics and Regularization

Temporal dynamics. To further analyze the temporal characteristics of the generated motions, Fig. A9 compares the speed profiles of a **ZeroCOI**-generated walking sequence with a ground-truth reference counterpart, including the root and two front paws. The generated motion exhibits a periodic alternating pattern between the left and right paws and deceleration-acceleration transitions around low-speed stance phases, both consistent with the reference walking dynamics. Meanwhile, the stable root speed and regular intervals between successive paw-speed peaks indicate a coherent gait rhythm throughout the sequence.

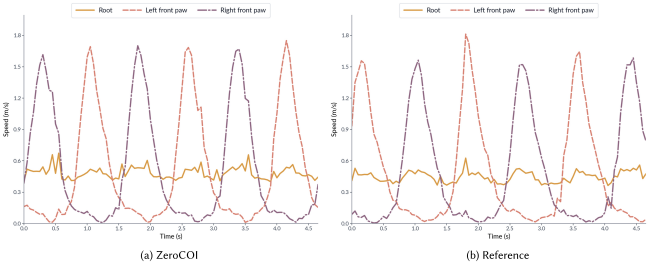


Fig. A9. Walking speed profiles of the root and two front paws for the **ZeroCOI**-generated motion and a ground-truth reference counterpart.

Temporal regularization. Existing physical constraints mainly focus on frame-level geometric validity. To further improve the temporal consistency, we additionally incorporate lightweight temporal regularization terms during training.

To reduce foot sliding, we penalize foot displacement across consecutive contact frames. Let $\mathbf{p}_{t,m}$ denote the world-space position of foot joint m at frame t, and let $c^f_{t,m} \in \{0,1\}$ denote its binary contact label. The sliding loss is defined as

$$\mathcal{L}_{\text{slide}} = \sum_{t,m} c^f_{t,m} c^f_{t+1,m} \left\| \mathbf{p}_{t+1,m} - \mathbf{p}_{t,m} \right\|_2^2 . \tag{A2}$$

To reduce persistent body-object penetration, we penalize consecutive frame penetration of the same body vertex, complementing the frame-wise constraint in Eq. (6). The temporal penetration loss is defined as

$$\mathcal{L}_{\text{temp-pen}} = \sum_{t,j} \max\left(0, -\text{SDF}(\mathbf{v}_{t,j})\right) \max\left(0, -\text{SDF}(\mathbf{v}_{t+1,j})\right) . \tag{A3}$$

Tab. A6 reports the quantitative results after incorporating these temporal regularizers. The additional constraints further reduce penetration and foot sliding while maintaining comparable performance on the other metrics. These results suggest that the physical plausibility of **ZeroCOI** can be further improved by incorporating targeted constraints.

E.5 Loss Sensitivity and Joint Convergence

Loss sensitivity. We further analyze the sensitivity to the contact and penetration loss weights by scaling their original weights with ratios of 1:1, 1:2, and 2:1. Tab. A7 reports the quantitative results, the model remains stable under these variations and exhibits an interpretable contact-penetration trade-off, with the default 1:1 setting achieving the best overall balance.

Joint convergence. During object-aware finetuning, video-domain SDS and explicit physical constraints are optimized jointly with empirically selected weights. Fig. A10 shows the training curves of the weighted video-domain SDS loss, physical losses, and total objective. Both supervision terms decrease steadily throughout training, while the total loss exhibits a consistent convergence trend.

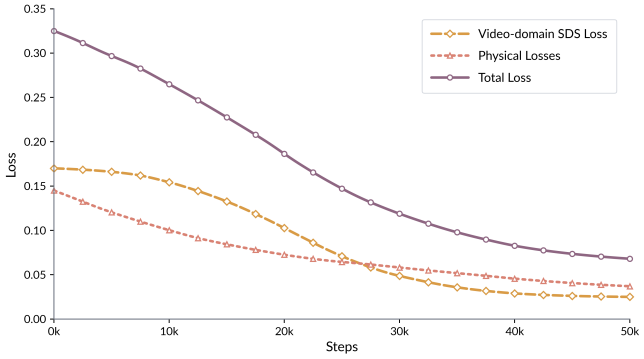


Fig. A10. Training curves of the weighted video-domain SDS loss, physical losses, and total objective loss during object-aware finetuning.

E.6 User Study

We conduct a user study with 35 participants to evaluate the perceptual quality of the generated interactions. Participants are asked to rate each video on a scale of 1 (worst) to 5 (best) along three dimensions: *Quality*, assessing overall visual fidelity; *Action*, measuring motion–text alignment; and *Physical Plausibility*, evaluating the realism of object interactions. Each participant evaluates 10 groups of videos, where each group contains results from all compared methods (including our method and baselines) under the same text prompt.

Tab. A8 reports the average scores across all participants. Our method achieves the highest scores on all three dimensions, demonstrating its effectiveness in producing visually realistic, semantically aligned, and physically plausible interactions.

E.7 Failure Cases

Fig. A11 illustrates typical failure modes of our method. One such failure mode arises in interactions with **extreme object geometries** (e.g., very thin or highly irregular shapes), where the generated motions may fail to establish stable contacts or accurately align interaction regions with the object surface. Another occurs for **complex**

Method	CLIP Score↑	CLIP Consistency↑	MD↑	MD_{local} ↑	$Pene_{\%obj}$ ↓	$Pene_{mean}$ ↓	$Pene_{max}$ ↓	FS↓
ZeroCOI	$\mathbf{28.2139^{\pm.169}}$	$0.9886^{\pm.000}$	$\mathbf{0.3472^{\pm.036}}$	$0.2546^{\pm.012}$	$0.0535^{\pm.013}$	$0.0218^{\pm.001}$	$0.0523^{\pm.002}$	$0.0439^{\pm.009}$
ZeroCOI + Temporal Reg.	$28.1987^{\pm.153}$	$\mathbf{0.9887^{\pm.001}}$	$0.3454^{\pm.029}$	$\mathbf{0.2579^{\pm.014}}$	$\mathbf{0.0478^{\pm.011}}$	$\mathbf{0.0194^{\pm.002}}$	$\mathbf{0.0471^{\pm.002}}$	$\mathbf{0.0325^{\pm.006}}$

Table A6. **Effect of temporal physical regularization.** Temporal regularization improves penetration and foot-sliding metrics while maintaining comparable performance on the other metrics.

Table A7. **Sensitivity analysis of contact and penetration loss weights.** C:P denotes the scaling ratio applied to the original contact and penetration loss weights.

C:P	$Pene_{\%obj}$ ↓	$Pene_{mean}$ ↓	$Pene_{max}$ ↓	FS↓	Contact↓
1:1	$0.0535^{\pm.013}$	$0.0218^{\pm.001}$	$0.0523^{\pm.002}$	$\mathbf{0.0439^{\pm.009}}$	$0.0291^{\pm.004}$
1:2	$\mathbf{0.0478^{\pm.011}}$	$\mathbf{0.0189^{\pm.002}}$	$\mathbf{0.0471^{\pm.003}}$	$0.0478^{\pm.010}$	$0.0367^{\pm.005}$
2:1	$0.0679^{\pm.015}$	$0.0305^{\pm.004}$	$0.0684^{\pm.006}$	$0.0465^{\pm.007}$	$\mathbf{0.0258^{\pm.004}}$

Table A8. **User study results.**

Method	Quality ↑	Action ↑	Physical ↑
DAViD	2.24	2.10	3.15
TSTMotion	2.67	3.27	2.02
AnimateAnyMesh	1.56	1.93	1.85
ZeroCOI (Ours)	**4.19**	**4.35**	**4.27**

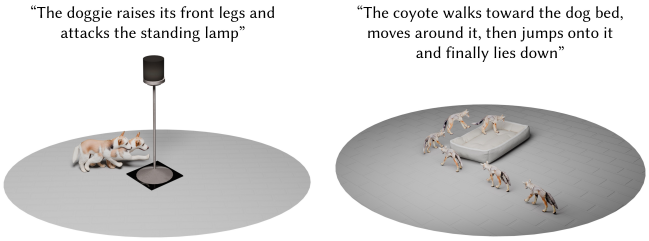


Fig. A11. **Failure cases.** Our method fails while facing extreme object geometries or too complex language descriptions.

language descriptions (e.g., long or compositional instructions involving multiple actions), where the generated behaviors may only partially satisfy the intended semantics of the prompt.

F Related Work

Animal Motion Generation & Dataset. Animal motion generation is more challenging than human motion generation due to data scarcity and skeletal diversity. Prior works leverage internet videos to circumvent the need for motion capture data, including reconstruction-based supervision [Cheng et al. 2023; Yang et al. 2025] and 2D weak constraints [Kapon et al. 2024; Sun et al. 2024], while typically relying on species-specific representations, limiting scalability across species and the ability to learn unified motion priors. To alleviate data scarcity, some works [Raab et al. 2024; Wang et al. 2025a; Zhang et al. 2024] explore one-to-many motion synthesis as a data augmentation strategy. In parallel, cross-category motion transfer methods align existing source trajectories from humans [Li et al. 2024] or other species [Zhang et al. 2025b; Zhao et al. 2024] across heterogeneous morphologies, rather than generating motions from scratch; following a similar spirit, OmniMotionGPT [Yang et al. 2024] transfers human motion priors to the animal domain to mitigate data scarcity. To address skeletal diversity, motion retargeting methods [Aberman et al. 2020; Chen et al. 2025c; Hu et al. 2023] map motions across characters with different skeletal topologies or body proportions while preserving source motion semantics under morphology changes. More recent efforts introduce larger-scale animal motion datasets and improve generalization across species, either through species-aware architectures [Wang et al. 2025c], unified cross-species representations [Wang et al. 2026], or topology-agnostic formulations [Chen et al. 2025b; Gat et al. 2025; Lee et al. 2025]. Despite these advances, existing methods primarily focus on isolated free-space motion and do not explicitly model interactions with 3D objects. This work targets the unified modeling of *text, objects, and canine motion*, enabling text-guided synthesis of Canine-Object Interactions (COIs) without paired interaction data.

Human-Object Interaction Generation. Benefiting from the availability of interaction datasets [Jiang et al. 2024a,b; Liu et al. 2025; Xu et al. 2025], data-driven methods model human interactions with static scenes [Hwang et al. 2025; Wang et al. 2024] or dynamic objects [Xu et al. 2023; Zeng et al. 2025], often under language conditions [Song et al. 2024; Wang et al. 2025b]. Subsequent works further improve interaction quality by exploiting fine-grained hand–object interactions [Cha et al. 2024; Ji et al. 2025; Paschalidis et al. 2025] or through physics-based tracking [Tevet et al. 2025; Wu et al. 2025; Yuan et al. 2023]. Despite their success, these approaches fundamentally rely on limited paired Human-Object Interaction (HOI) data, limiting their scalability. To address this issue, recent studies explore zero-shot HOI synthesis by leveraging external priors. Some methods [Kim et al. 2025, 2024; Li et al. 2026; Li and Dai 2024] use pre-trained image or video diffusion models, first generating 2D HOI observations and then lifting them to 3D/4D representations. This process is typically coupled with iterative optimization and geometric constraints to enhance physical plausibility and mitigate depth–scale ambiguity. Others [Cao et al. 2025; Dai et al. 2024; Dai and Xu 2026; Zhu et al. 2025] adopt Score Distillation Sampling (SDS) to optimize HOI directly in the 3D/4D space. Some studies [Gui et al. 2026; Guo et al. 2025; Xu et al. 2024] introduce LLMs to provide high-level interaction semantics, contact relationships, or action intents, while more recent works [Li and Dai 2026; Lou et al. 2025; Mu et al. 2026; Zhang et al. 2025a] jointly leverages language and visual diffusion priors to impose multimodal constraints for HOI generation. However, these zero-shot HOI methods often rely on instance-specific optimization or inference-time adaptation, limiting

scalability and preventing the reuse of interaction knowledge across objects, scenes, or prompts. We focus on zero-shot COI generation and adopt a scalable formulation that learns reusable interaction knowledge during training. Our approach leverages visual diffusion priors as the interaction supervision, avoiding paired interaction data and instance-specific optimization, while facilitating generalization.

Visual Diffusion Priors for 4D Generation. Recent advances in visual diffusion models [Blattmann et al. 2023; Rombach et al. 2022; Ruiz et al. 2023] have enabled high-quality, controllable image and video generation with multimodal conditioning. Building on their capabilities, Score Distillation Sampling (SDS) [Poole et al. 2023] has been widely adopted for 3D/4D content generation [Bahmani et al. 2024; Ren et al. 2023; Singer et al. 2023], leveraging gradients from pre-trained visual diffusion models to guide the optimization of NeRFs [Mildenhall et al. 2020] or Gaussian Splatting [Kerbl et al. 2023]. ProlificDreamer [Wang et al. 2023] further generalizes SDS to Variational Score Distillation (VSD), modeling 3D scene parameters as a distribution instead of a single deterministic representation, while remaining a per-scene optimization framework. Beyond single-object settings, SDS has been extended to 4D compositional scene generation [Sun et al. 2025a; Yu et al. 2024], modeling dynamic interactions among multiple objects. In parallel, visual diffusion models have been employed to indirectly facilitate 4D reconstruction [Mou et al. 2024; Shao et al. 2024; Sun et al. 2025b]. Recent studies further show that video diffusion models contain decodable physical-plausibility signals in their internal states [Esmati et al. 2026] and can serve as latent world simulators for 3D scene understanding [Wu et al. 2026]. In this work, we leverage visual diffusion priors via an SDS-based formulation, using them as implicit semantic and physical preference signals to guide the motion generator during training, rather than optimizing individual 4D instances.

Distribution Matching Distillation. Distribution Matching Distillation (DMD) [Yin et al. 2024b] distills a multi-step diffusion model into an efficient few-step generator by aligning the student and teacher distribution using diffusion score signals. Follow-up works improve this paradigm from different aspects, including training stability [Yin et al. 2024a], efficient convergence [Chen et al. 2025a], and extensions to flow-based models [Ge et al. 2026]. Adversarial distillation [Lin et al. 2024; Sauer et al. 2024a,b] provides another related distribution-matching formulation, using an explicit discriminator to match the target distribution. Our distribution-level video-domain SDS shares a similar supervision mechanism with these works, in that both leverage score signals from pre-trained diffusion models to guide a parametric generator. However, existing DMD methods mainly aim to reduce diffusion sampling steps while approximating the same visual distribution as the teacher, whereas we use video diffusion scores as cross-domain semantic supervision to train a 3D motion generator for zero-shot canine–object interaction synthesis.

G Limitations

Video Prior Biases. Despite mitigating data scarcity in the canine motion domain by leveraging large-scale pre-trained text-to-video diffusion models, **ZeroCOI** inevitably inherits the biases and coverage limitations of the underlying video priors, which may result in suboptimal behaviors for rare actions, less common interaction patterns, or highly specialized physical constraints. Stronger DiT-based video priors may provide richer visual interaction supervision, but their substantially larger parameter counts and spatio-temporal attention incur higher memory consumption, while slower denoising inference reduces training efficiency when used as training-time priors. Extending **ZeroCOI** to such stronger video priors therefore requires better balancing supervision quality with computational efficiency.

Residual Physical Artifacts. The physical fidelity of **ZeroCOI** is partly bounded by the quality of the source motion datasets used to construct DogML3D. These datasets are primarily obtained from synthetic or game-extraction pipelines rather than high-fidelity motion capture, and may therefore contain physical artifacts such as foot sliding or imperfect contact. In addition, mapping heterogeneous source skeletons to a unified SMAL representation through retargeting may introduce additional approximation errors. As analyzed in Sec. E.2, **ZeroCOI** reduces such artifacts to some extent, but further improving motion naturalness still requires higher-quality motion priors.

Reliance on Synthetic Interaction Descriptions. Due to the absence of paired text-object-motion data, we synthetically augment motion-centric descriptions into interaction-centric instructions to provide object-aware interaction semantics during finetuning. However, these descriptions are generated by an LLM under a predefined augmentation scheme. Compared with manually annotated interaction descriptions, they may exhibit reduced linguistic diversity and more limited coverage of fine-grained interaction semantics.

Lack of Dynamic Simulation. While **ZeroCOI** encourages physical plausibility through lightweight 3D constraints, it does not explicitly enforce dynamics-level physical realism through simulation-based training or refinement. Incorporating full physics simulation would require an actuated canine agent model with reliable physical parameters, which is difficult to obtain across diverse canine morphologies. Moreover, coupling a simulator with diffusion model training would substantially increase system complexity and training cost. Therefore, achieving higher-fidelity physical realism remains challenging under the current physical-modeling and computational constraints.

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